



CSE-476: Introduction to Natural Language Processing
Quiz 1

Multiple Choice Questions (Single correct)

Instructions: Select the correct option with a short explanation.

- Q1. In the BIO tagging scheme for NER, what does the “B-” prefix indicate?
- (A) Beginning of an entity
 - (B) Inside an entity
 - (C) Outside any entity
 - (D) Between two tokens
- Q2. Which statement best describes a (plain) Markov chain in sequence tagging?
- (A) It uses only words (observations) and ignores labels
 - (B) It is identical to an HMM
 - (C) It models transitions between labels but does not consider the words
 - (D) It models emissions $P(\text{word} \mid \text{label})$
- Q3. In a Hidden Markov Model (HMM) for NER, an emission probability is:
- (A) $P(\text{label} \mid \text{previous label})$
 - (B) $P(\text{label} \mid \text{word})$
 - (C) $P(\text{word} \mid \text{previous word})$
 - (D) $P(\text{word} \mid \text{label})$
- Q4. Which of the following best describes Semantic Role Labeling (SRL)?:
- (A) A task that identifies predicates in a sentence and labels roles of elements such as agent, theme, or location
 - (B) A process that assigns grammatical roles like subject and object based on syntactic structure
 - (C) A method that classifies named entities and links them to a knowledge base
 - (D) A technique that labels words with their parts of speech and morphological features

- Q5. Which of the following best describes Byte Pair Encoding (BPE)?
- (A) A technique that compresses text by replacing consecutive repeated characters with a count and a symbol
 - (B) A tokenization method that iteratively merges the most frequent pair of symbols in a corpus
 - (C) A tokenization method that splits text based on whitespace and punctuation
 - (D) A character-level encoding scheme that assigns fixed-length binary codes to characters
- Q6. Which of the following best describes the main goal of tokenization in NLP?
- (A) To label each token with a grammatical category
 - (B) To convert text into a sequence of symbols suitable for modeling
 - (C) To reduce vocabulary size to zero
 - (D) To assign probabilities to sentences
- Q7. Why is splitting text by spaces (`text.split(" ")`) considered an insufficient tokenizer?
- (A) It produces too many tokens
 - (B) It does not work well for languages without explicit word boundaries
 - (C) It cannot tokenize English at all
 - (D) It ignores punctuation entirely
- Q8. Which of the following is NOT listed as a common sequence tagging task?
- (A) Part-of-speech tagging
 - (B) Named Entity Recognition
 - (C) Dependency parsing
 - (D) Sentiment analysis
- Q9. What is a common limitation of BIO tagging for Named Entity Recognition?
- (A) It cannot represent overlapping or nested entities
 - (B) It requires too many labels
 - (C) It does not support multiple entity types
 - (D) It ignores entity boundaries

- Q10. When choosing between character-level tokenization and sentence-level tokenization, which trade-off must be considered?
- (A) Speed vs accuracy
 - (B) Tokenized sequence length vs vocabulary size
 - (C) Precision vs recall
 - (D) Memory vs computation

End of Paper

- Q9. Which issue involves a tagging scheme that requires an excessively large number of labels?
- A.** It cannot represent overlapping or nested entities
 - B.** It requires too many labels
 - C.** It does not support multiple entity types
 - D.** It ignores entity boundaries
- Q10. In model selection, which trade-off refers to balancing system speed and accuracy?
- A.** Speed vs accuracy
 - B.** Tokenized sequence length vs vocabulary size
 - C.** Precision vs recall
 - D.** Memory vs computation